

Detailed Point-by-Point Revision Response

Draft working document — Manuscript 3400076, Advanced Intelligent Systems

"Internal Camera-based Reconstruction and Closed-loop Control of Soft Robotic Arms"

Editor: Dr. Richard Murray | Deadline: October 13, 2026 | Reviewer 1 only

This document works through every comment using the manuscript's own content, data, and reported statistics wherever they allow a substantive answer. Two constraints shape it throughout: (1) no new physical experiments are run — only re-analysis of data already collected and reported; (2) where a point genuinely depends on information outside the manuscript (raw logs, hardware specs, code, the five cited comparison papers in full), this is flagged explicitly as needing your input rather than guessed at.

Editor's General Requirements

G1. Title precision. Current title ("Internal Camera-based Reconstruction and Closed-loop Control of Soft Robotic Arms") already names the method, sensing modality, and system; it reads as adequately keyword-searchable. No change proposed unless you want one.

G2. Reviewer-suggested references. The five suggested papers (Hofer 2021, Thuruthel 2017, Han 2025, Lu 2024, Yoo et al.) are addressed in Comment 2 below; all five are incorporated into the related-work discussion, so no need to decline any of them in the response letter.

G3. Proofread / acronym audit. Full pass needed. Acronyms used without a spelled-out first occurrence to check: PCC, PPC, CNN, MLP, GAP, TPU, PLA, FBG, IMU, RRT, SLERP, IK, FK. The abstract and title currently avoid acronyms entirely, which already satisfies the standalone-text requirement for those two; the ToC/graphical abstract text (if one exists separately) needs the same check.

G4/G5. Figure text, spelling, captions. Folded into Comment 10 (figure redesign) and the caption checks under Minor Comments 21-27.

G6. Copyright / image permissions. All figures appear to be original renders/schematics/plots produced for this manuscript (robot photos, CNN/MLP diagrams, PCC schematics, data plots) rather than reproduced from other sources. Recommend stating this directly in the response letter rather than seeking permissions.

G7. Reference format (full titles, DOIs). Reference list currently gives full article titles already (see References 1-36); DOIs are largely absent and should be added where available.

Major Comments

Comment 1: Role of visual sensing in closed-loop control

Reviewer's point: Unclear whether curvature feedback enters control; Algorithm 1's PCC-based tendon-length initialization appears unused once the loop starts.

Response / substance:

Checking Algorithms 1 and 2 against the text confirms the reviewer's reading exactly. In Algorithm 1, camera feedback $f_p^3(l)$ supplies only the current pose p used in the task-space error ($p_{ref} - p$); the Jacobian J and the null-space term $N \cdot (l_{ref} - l)$ are both computed from tendon lengths l read from the motor encoders, not from vision. Algorithm 2 (data-driven) uses only camera pose, with no encoder input at all, consistent with the text's claim that it "avoids the use of encoders." The curvature networks f_C^1 and f_C^3 never appear in either algorithm; they are evaluated only as a separate reconstruction capability (Sections 4.1.2 and 4.2.2). This three-way separation (camera pose for the task error, encoders for the Jacobian/null-space, curvature reconstruction as a standalone output) is correct in the implementation but not stated anywhere as a single, explicit sentence, which is what caused the ambiguity. On the flagged Algorithm 1 line: it computes an initial tendon-length estimate from the camera-derived tip positions via the PCC inverse map (Eq. 2) before the loop begins, but the very first loop iteration immediately overwrites l with an encoder reading, so this estimate never reaches J , N , or δl . Its only plausible remaining function is a pre-motion consistency check between the camera- and encoder-based state estimates (e.g., to catch a gross camera miscalibration before moving the motors) — but the manuscript does not currently claim this. This needs a decision.

Manuscript changes:

- Abstract: add one clause distinguishing pose-based control feedback from curvature-based reconstruction.
- Introduction, contribution items 2-3: state explicitly which signal (camera pose vs. encoder tendon length vs. curvature) feeds which part of the control law.
- Fig. 4c caption: annotate that camera feedback enters only the pose/error path; encoders feed the Jacobian and null-space term.
- Algorithm 1: either remove the flagged pre-loop line, or keep it with one sentence describing it as a pre-motion consistency check (see author input below).
- Conclusion (Section 5): restate the same three-way distinction.

Needs author input (not derivable from the manuscript alone): Was the camera-vs-encoder pre-loop comparison in Algorithm 1 actually monitored/used as a safety check during your experiments (e.g., logged and inspected, or used to abort a run)? If yes, keep the line and we can justify it with that evidence; if it was just a leftover from an earlier version of the algorithm, the cleanest fix is to delete it.

Comment 2: Originality relative to prior work

Reviewer's point: Individual sensing/control elements are not novel in isolation; originality must be established through direct comparison with named prior systems.

Response / substance:

Using the reviewer's own one-line descriptions of the five suggested papers (Yoo et al. is already cited as Ref. [15] in the current manuscript, for zero-shot sim-to-real internal-camera proprioception), the distinguishing angle for this paper is consistent across all five comparisons: this is, to our knowledge, the only system combining (i) distributed markerless internal cameras across a modular multi-section body, with (ii) both a model-based and a learned controller that explicitly exploit kinematic redundancy through a null-space secondary objective, validated on both single- and multi-section hardware. Hofer et al. (2021) reaches higher single-actuator accuracy ($\sim 1^\circ$ at 30 Hz vs. our $3.11^\circ \pm 1.96^\circ$ at 8 Hz) but on a single spherical actuator with no redundancy to exploit. Thuruthel et al. (2017) is the closest in spirit to our data-driven controller (learned inverse kinematics with feedback on a redundant tendon-driven arm, tested under load) — the distinguishing claim here would be the vision-based, encoder-free sensing pipeline and the explicit redundancy-weighted

training loss (Eq. 11), but this must be checked against their actual feedback modality before writing it into the paper. Han et al. (2025) also uses an internal camera for full-body deformation and force estimation; the distinguishing claim is our closed-loop null-space control (which requires deciding whether their internal-camera system is used for control at all, or sensing only), and conversely their force-sensing capability is something we explicitly lack and already list as a limitation (Section 5). Lu et al. (2024) is FBG-based, not vision-based, and belongs in the related-work sensing discussion (Section 1, paragraph 2) rather than as a direct comparator. Yoo et al. targets sim-to-real transfer for a different actuation type; the distinguishing point is that our networks are trained on real, directly-collected data and are paired with closed-loop null-space control, which their work does not address.

Manuscript changes:

- Add a dedicated comparison paragraph in the Introduction, after the sentence ending "...closed-loop null-space control of continuum robots, either through model-based or learning-based approaches."
- Sharpen the contribution list (Section 1, items 1-4) to state the integration claim precisely: distributed markerless internal-camera sensing + redundancy-aware control (model-based and learned) on a multisection tendon-driven arm.

Needs author input (not derivable from the manuscript alone): This comparison is currently built only from the reviewer's one-sentence summaries. Each of the five papers should be read in full before this paragraph is finalized, in particular to confirm: Thuruthel et al.'s actual sensing modality (vision, motion capture, or embedded sensors), and whether Han et al. or Hofer et al. include any redundancy/null-space control. An inaccurate comparison is worse than none.

Comment 3: Scalability claim overstated

Reviewer's point: The estimator is not invariant to section count; the output head changes (6→18 pose variables, 1→3 curvature parameter sets), not only the first convolutional layer, and only $n=1/n=3$ were tested.

Response / substance:

This is a direct factual correction to accept. Section 3.1 currently states that scaling to a different section count requires "adjusting only the first layer," but Fig. 3a and the text's own description of the regression head (6 outputs for single-section, 18 for multi-section) confirm the output head also changes. The claim should be narrowed to what is actually demonstrated: a modular, transfer-learning-based architecture where the pretrained MobileNetV2 backbone is reused across section counts, with both the first convolutional layer (input channels) and the final regression head (output width) adapted and retrained for each new n . This has been validated only at $n=1$ and $n=3$; $n=2$ and $n>3$ were neither trained nor tested, and no claim of zero-shot generalization to an untested section count should remain.

Manuscript changes:

- Section 1, contribution item 1: rewrite "scales to arms with different numbers of sections" to state retraining is required, with the backbone reused via transfer learning.
- Section 3.1, sentence describing Fig. 3a: correct "requires adjusting only the first layer" to include the output head.
- Add one explicit sentence noting $n=1$ and $n=3$ are the only validated configurations.

Comment 4: Multi-section curvature ground truth is a learned surrogate

Reviewer's point: SI Section 4.2 states marker occlusion prevents simultaneous full-body tracking on the multi-section arm; a learned network N_C maps tip pose to curvature labels instead, so the multi-section CNN is evaluated against a model-constrained surrogate, not direct measurements.

Response / substance:

The reviewer's characterization matches the SI description. This affects Section 4.2.2, Fig. 7b, and part of Fig. 8a. Two things are fully addressable without new experiments: (1) reframing the result honestly, and (2) reporting N_C 's own validation accuracy numerically instead of only through Fig. S2b's plot. One thing is not addressable without new experiments: a direct backbone-vs-marker comparison, since the underlying occlusion problem that prevented that measurement during data collection cannot be fixed retroactively.

Manuscript changes:

- Reframe Section 4.2.2 and the relevant Fig. 8a discussion as "model-constrained reconstruction," not validated backbone reconstruction, for the multi-section arm.
- State the underlying assumptions explicitly near Eq. 5: planar bending within each section, no torsion (already implied by the existing gimbal-lock sentence), and constant bending direction ϕ within a section under the PPC model used here.
- Report N_C 's numerical validation error (mean/median, sample count, interval type) replacing the purely graphical Fig. S2b, and state whether its validation set is disjoint from the data used to train the three-section curvature CNN.
- Add one explicit sentence (Section 2.2 or 4.2.1) stating that direct backbone-vs-marker validation for the multi-section arm was not possible due to inter-section marker occlusion during data collection, and that resolving this would require a new capture session with an occlusion-avoiding marker layout, which is outside this revision.

Needs author input (not derivable from the manuscript alone): The exact N_C validation numbers (error statistics, validation sample count, whether the interval shown in Fig. S2b is std, SEM, or another measure) need to be pulled from your training logs; I don't have access to those artifacts.

Comment 5: Curvature-coordinate inconsistency (normalized vs. dimensional arc length)

Reviewer's point: Main-text Eq. 4 uses normalized arc length $s \in [0,1]$; SI Eqs. S5-S9 evaluate the polynomial at $L/3, 2L/3, L$ and use $c_0 = \theta/L$, implying dimensional arc length. These cannot both be correct as written.

Response / substance:

This is a genuine, fixable inconsistency; it does not require new data, only consistent rederivation. The SI equations and the closed-form initialization already operate in dimensional arc length $x \in [0, L]$ (using $c_0 = \theta/L$, which only has correct units if the independent variable is dimensional length), so the cleanest fix is to convert the main-text formulation to match the SI rather than the reverse, since that preserves the existing fitting code and initialization equations unchanged.

Manuscript changes:

- Eq. 4: rewrite as $\theta(x) = c_0 \cdot x + (1/2)c_1 \cdot x^2 + (1/3)c_2 \cdot x^3$, $x \in [0, L]$, with c_0, c_1, c_2 now carrying units of curvature and its derivatives ($1/\text{length}$, $1/\text{length}^2$, $1/\text{length}^3$), rather than the dimensionless coefficients implied by normalized s .
- Eqs. 5a-5b: replace the normalized- s integration variable with x directly; remove the explicit extra factor of L inside the translation integral (Eq. 5b), since x already carries the length dimension.
- Eq. 6 (least-squares fit): the evaluation points $s = 1/3, 2/3, 1$ become $x = L/3, 2L/3, L$, which then matches SI Eqs. S5-S9 exactly and removes the inconsistency at its source.
- Update nomenclature/notation list (main text and SI) to state units for c_0, c_1, c_2 explicitly.

Comment 6: Secondary objective not quantitatively evaluated

Reviewer's point: No experiment isolates the effect of the null-space secondary objective $H(l)$; no reported tendon-limit margins, tendon distributions, or with/without comparison; $H(l)$ penalizes deviation from mean tendon length, not force/current/energy directly.

Response / substance:

Two levels of response are possible here without new hardware trials. First, a retrospective quantitative report: tendon lengths were logged via motor encoders at every control iteration in both the target-reaching (Sections 4.1.4, 4.2.4) and trajectory-tracking (Sections 4.1.5, 4.2.5) experiments already run. $H(l)$ (or equivalently the RMS deviation of each tendon from l_{mean}) can be computed directly from those logs for both controllers at convergence on all 20 k-medoids targets and along the tracked trajectories, giving exactly the "tendon-limit margins, tendon distributions" the reviewer asks for, as a re-analysis rather than a new experiment. Second, a genuine with/without ablation: using the already-trained forward-kinematics network

N_p^3 (data-driven case) or the PCC forward map (model-based case), the resolved-motion-rate law (Eq. 8/10) can be re-run offline with the null-space term $N \cdot (I - I)$ set to zero, driving toward the same 20 targets, and the resulting tendon distributions compared against the logged real runs with the secondary objective active. This isolates the objective's effect computationally; it should be labeled clearly as an offline/simulated comparison, not a new hardware ablation. On framing: $H(I)$ is a quadratic penalty on deviation from the mean sampled tendon vector, which is correctly described as actuator-range centering; it does not directly minimize force, current, or energy unless those were separately measured.

Manuscript changes:

- Add a new figure/table (Section 4.1.4/4.2.4 or SI) reporting achieved $H(I)$ / tendon-margin distributions from existing closed-loop logs, for both controllers.
- If the offline zero-null-space simulation is run, add it as a companion panel with an explicit "simulated, not measured on hardware" label.
- Revise the Eq. 9 description to state plainly that $H(I)$ is an actuator-range-centering objective, and remove or qualify any language implying force/energy minimization unless motor current data substantiates it.

Needs author input (not derivable from the manuscript alone): Is motor current/torque telemetry logged from the DYNAMIXEL XL330-M288-T motors during the closed-loop runs? If so, an energy-proxy claim becomes possible instead of only actuator-range centering; if not, the actuator-range-centering framing is the honest one to keep.

Comment 7: Control evaluation is narrow (coverage, repeatability, disturbance generality)

Reviewer's point: Bending angles in the control evaluation are much smaller than the training range (up to 85°); 20 targets are visited once each; 30 trajectory waypoints are serially correlated, not independent trials; the 50 g disturbance test covers only one load condition.

Response / substance:

This comment has one fully re-analyzable part and two parts that genuinely require new physical trials. Coverage quantification is re-analyzable: the 20 k-medoids targets and the tracked trajectory waypoints already have logged poses (and, via the PCC/curvature networks, corresponding bending angles), so the actual bending-angle and compression range spanned by the control evaluation, and its fraction of the sampled workspace, can be computed directly and reported numerically, addressing the Figs. S4-S6 coordinate-reporting request. Repeatability and broader/nonplanar coverage and additional disturbance conditions are not re-analyzable: each target and trajectory was run once, so there is no repeated-trial data to compute variance from, and no nonplanar/boundary-path/additional-load data exists to report. For these, the honest path is to narrow the claims rather than imply data that isn't there: state plainly that reported standard deviations in Figs. 5c and 8b describe variation across different targets and waypoints within a single run, not trial-to-trial repeatability; state the disturbance result as a single-condition demonstration rather than general robustness; and list repeated-trial statistics, nonplanar trajectories, and multi-condition disturbance testing explicitly as future work in Section 5, which already has a limitations paragraph this fits naturally into.

Manuscript changes:

- Add a coverage-quantification subsection or SI table: bending-angle/compression range of the 20 targets and tracked trajectories vs. the training dataset's sampled range, computed from existing logs.
- Sections 4.1.4/4.2.4 and 4.1.5/4.2.5: reword standard-deviation interpretation (across-target/waypoint variation, not repeatability).
- Section 4.2.6: reframe the 50 g payload result as a single-condition demonstration, not general disturbance robustness.
- Section 5 (Limitations): add repeated-trial evaluation, nonplanar/boundary-path trajectories, and multi-magnitude/direction disturbance testing to the future-work list.

Comment 8: Train/test independence not established

Reviewer's point: 85/10/5 split is described, but acquisition sequence, session structure, and whether temporally adjacent configurations could land in both training and test sets are not documented.

Response / substance:

This is a documentation gap that depends entirely on how the split was actually implemented, which is not recoverable from the manuscript text; it needs to come from the data pipeline itself.

Manuscript changes:

- Add a paragraph to Section 3.1.1 or the SI stating exactly how the 85/10/5 split was performed (uniform random over all samples, or blocked by acquisition session/trial).
- If the split was not session-blocked, state this plainly as a limitation, since temporally adjacent frames from the motor-babbling and structured sampling procedures (Section 3.2) may be correlated.
- State explicitly whether the 20 k-medoids control targets and trajectory waypoints were excluded from network training and validation.

Needs author input (not derivable from the manuscript alone): How exactly was the 85/10/5 split implemented (random shuffle vs. session-blocked), and were the control-evaluation targets/trajectories excluded from the training data? This has to come from your actual data-splitting code or lab notes.

Comment 9: Missing reproducibility details

Reviewer's point: Camera model/lens/exposure, training hyperparameters (epochs, seeds, normalization), and robot/tendon specs (material, diameter, preload criterion, number of sessions) are not reported.

Response / substance:

The manuscript already reports a substantial amount of this without a consolidated table: NVIDIA RTX 3090 GPU; Adam optimizer, $\eta=10^{-3}$, weight decay, early stopping, Huber loss for all regressors; MobileNetV2 pretrained backbone with transfer learning (f_C^1 initialized from f_p^1 ; f_p^3 initialized from f_p^1); 85/10/5 split; input images 480×640×n grayscale; three 150 mm, 60 mm-diameter Trimmed Helicoid modules; DYNAMIXEL XL330-M288-T motors; 6 mm pulley wheels; PLA base/pulleys/camera supports, TPU helicoidal components; $\lambda=0.1$; $\gamma=3$ mm (single-section) / 10 mm (multi-section); $\epsilon=1$ mm waypoint spacing; $\zeta=0.01$; six OptiTrack Prime 13 cameras with Motive software; 10,000 data points per arm. Consolidating these into one SI table already closes most of the gap. What remains missing cannot be pulled from the text and needs to come from you directly.

Manuscript changes:

- Add a single SI reproducibility table consolidating all parameters already reported in the main text (listed above), plus the items flagged below once supplied.

Needs author input (not derivable from the manuscript alone): Camera manufacturer/model and lens/field-of-view for the embedded miniature cameras; exposure/gain behavior; maximum training epochs and early-stopping patience; random seed(s); output normalization statistics for pose/curvature targets; tendon material and diameter; numeric current-limit value used in position-control mode; numeric preload/zeroing current threshold; number of distinct data-acquisition sessions. None of these are in the manuscript or recoverable by me from it.

Comment 10: Figure clarity (Figs. 1B, 2, 3, 5)

Reviewer's point: Visual density and inconsistent presentation make the methodology hard to follow; Fig. 3 in particular does not make the relationship between the pose, curvature, forward-kinematics, and inverse-kinematics networks clear, and tensor dimensions/network I/O are not labeled.

Response / substance:

Fig. 3 currently shows the CNN architecture (panel a) and MLP architecture (panel b) as two separate blocks without indicating which specific network instance ($f_p^{1/3}$, $f_C^{1/3}$, N_p^3 , $N_l^{1/3}$) flows through each path, and without stating tensor dimensions at each stage (e.g., 480×640×n input, 1280-channel GAP output, 128-dimensional MLP latent space already mentioned in the caption text but not shown on the diagram itself).

Manuscript changes:

- Redraw Fig. 3 with a single consistent flow direction, explicit tensor-dimension labels at each stage, and clear visual grouping distinguishing the four networks (pose, curvature, forward kinematics, inverse kinematics).
- Revisit Fig. 1B (robot/section schematic) and Fig. 2 (PCC parameterization) for label density.

- Revisit Fig. 5 panel layout for readability (ties into Minor Comments 22-23 on axis scaling and statistic definitions).

Minor Comments

11. pp. 1-2 — "Markerless" is imprecise: external optical markers generate the supervised training labels, even though no markers are needed at deployment.

Response: Replace with a phrase that distinguishes training-time supervision from deployment: e.g., "markerless at inference (external motion-capture markers are used only to generate training labels)." This is a wording fix throughout Sections 1-2.

12. pp. 1-2 — Inconsistent use of "section" vs. "segment."

Response: Standardize on "section," the dominant term in the body text and figures; update the Section 4.1/4.2 headers ("Single-Segment"/"Multi-Segment") and any remaining "segment" instances to match.

13. pp. 3-5 — Calibration paragraph states "all ten tendons" are tightened; the multi-section arm has nine (3 modules $\times$ 3) and the single-section arm has three.

Response: Correct the tendon count for each arm and state explicitly whether the same current-controlled preload procedure applies identically to both, given the single-section arm is directly actuated while Modules II/III of the multi-section arm are actuated via decoupled Bowden cables.

14. pp. 3-5 — Camera manufacturer, model, lens/field-of-view, frame rate, and physical size/mass are not given.

Response: Add to Section 2.1 or the new SI reproducibility table (see Comment 9).

15. pp. 3-5 — "Simultaneous" image acquisition across the three USB-connected cameras is claimed without a stated synchronization method; software-triggered capture over USB is not necessarily hardware-synchronous.

Response: Either specify the actual synchronization mechanism (hardware trigger vs. software timestamp matching), or soften "simultaneous" to "acquired in close temporal proximity" if only software-level synchronization was used.

16. pp. 5-7 — How sine/cosine angular outputs are decoded back to angles, and how position/orientation/length/curvature losses are balanced after normalization, is not explained.

Response: Add the decoding formula (atan2 of the predicted sine/cosine pair, presumably) and state the loss-weighting scheme (equal weights after normalization, or explicit per-term weights) in Section 3.1.1.

17. pp. 5-7 — The negative sign in SI Eq. S11 and the definition of the Grad-CAM-highlighted pixel regions are unclear.

Response: Add a sentence deriving/explaining the sign convention in Eq. S11 and explicitly define what the highlighted regions in Fig. 4a represent (gradient magnitude threshold, top-k activation, etc.).

18. pp. 5-7 — Duplicate of Comment 1 (unused PCC-init line in Algorithm 1).

Response: Same fix as Comment 1.

19. pp. 5-7 — Jacobian notation (J_{t2l} , J_{l2p} , and the learned inverse-Jacobian derivatives) is inconsistent between a forward Jacobian, its inverse, and a network derivative.

Response: Standardize notation: e.g., use J for the analytic (PCC-based) Jacobian with subscripts denoting input $\rightarrow$ output variables, and a distinct symbol (already partly used, e.g. with a hat) for the learned Jacobian estimated as a network derivative, applied consistently in Eqs. 7-10 and Algorithms 1-2.

20. pp. 5-7 — Whether the open-loop/feedforward baselines share the same initial pose, planner, gains, and stopping conditions as the closed-loop controllers is not stated; "open-loop" and "feedforward" are used inconsistently.

Response: State explicitly in Section 3.3 that the open-loop/feedforward baselines reuse the same planner (W^1/W^3), gains (λ , ϵ), and target/waypoint sequence as the closed-loop controllers, differing only in that state updates come from internal forward-kinematics rather than camera feedback. Use one term ("open-loop" is recommended, matching Table 2/Fig. 5c/8b) consistently.

21. pp. 8-11 — The denominator for every percentage error is not stated (section length vs. cumulative tip length vs. maximum deployed length).

Response: State the normalization basis explicitly at first use of each percentage figure (Sections 4.1.1, 4.1.4, 4.2.1, 4.2.4).

22. pp. 8-11 — Fig. 5b's nonuniform axis scaling visually distorts the reconstructed shapes.

Response: Add equal-scale axes or a second undistorted panel with explicit scale bars, alongside the current bending-emphasized view.

23. pp. 8-11 — Fig. 7 alternates between "cumulative," "sum," and "average" 3D error without a precise definition; summing 3 points (single-section) vs. 9 points (multi-section) prevents direct comparison.

Response: Define the plotted statistic precisely, and consider reporting per-point RMSE or error vs. normalized arc length instead of a summed value, so the two arms become directly comparable.

24. pp. 8-11 — Table 1 states orientation error "remains below 9°," but several reported standard deviations exceed their means (e.g., Section I: $6.78^\circ \pm 15.94^\circ$).

Response: Clarify that only the mean is below 9°, and add median, 95th percentile, and maximum error alongside mean/std in Table 1 — a direct re-analysis of the already-computed per-sample test errors, no new data needed.

25. pp. 8-11 — "Bent configurations improve accuracy" (Sections 4.1.1, 4.1.2) is currently a qualitative claim.

Response: Plot reconstruction error against bending angle using the existing per-sample test-set predictions (already computed for Figs. 5a-5b) to support or qualify the claim quantitatively; this is re-analysis of existing results, not new data collection.

26. pp. 8-11 — Figs. 6, 9, 10 do not state start direction, waypoint order, number of completed cycles, or whether plotted paths are a single run or an average, and lack quantitative path-error metrics.

Response: Add these details to each caption, and compute a quantitative path-error metric (e.g., RMS radial deviation and RMS height deviation from the reference circle) from the already-logged trajectory data, reported in the caption or an accompanying table.

27. pp. 8-11 — The Discussion (Section 5) does not clearly separate demonstrated capabilities from proposed future applications (obstacle avoidance, whole-body grasping, surgical guidance, force/contact sensing).

Response: Revise Section 5 to state explicitly that these are future directions, not capabilities evaluated in this manuscript; this is consistent with, and reinforces, the limitations paragraph already present there.

Supporting Information Comments

S28. Fig. S2b lacks numerical N_C error, validation sample count, and interval type (std/SEM/other).

Response: Report these numbers directly from the existing N_C validation run; this is the same data underlying the current plot, just reported as text/table in addition to the figure. See also Comment 4.

S29. Table S6 reports highly skewed errors (e.g., 11.92 ± 25.10 mm for Section II) using only mean $\pm$ std, which is misleading for a skewed distribution.

Response: Recompute and report median and interquartile range (or another robust summary) alongside mean/std for every entry in Table S6, and add a sentence discussing the outliers driving the large standard deviations. This is a re-analysis of already-collected per-target error data.

S30. Figs. S4-S6 lack matched target identifiers or consistent color scales across controllers, making direct visual comparison difficult.

Response: Add a consistent target/pair ID to each plotted point (the same 20 k-medoids targets are used across all four controllers, so this is a labeling/formatting fix, not new data) and standardize color-scale limits across the four subfigures.

S31. Tables S3-S4 list dataset fields but not sample counts per acquisition mode/robot configuration, including how many samples were removed and why.

Response: Add sample counts broken down by the three sampling regimes described in Section 3.2 (all-modules-same, two-coupled/one-independent, fully independent), and report any excluded/filtered samples with the reason, from the existing dataset records.

Summary of what still requires your input before this can become the final response letter: (1) full reading of the five comparison papers for Comment 2; (2) a decision on the flagged Algorithm 1 line for Comment 1; (3) confirmation of the data-split procedure for Comment 8; (4) the hardware/training parameters listed under Comment 9; (5) confirmation of whether motor-current telemetry exists for Comment 6's possible energy-proxy claim; (6) the N_C validation numbers for Comment 4/S28. Everything else in this document is either a direct textual/derivational fix or a re-analysis of data already reported in the manuscript and its SI.